TulipFleet – Map Implementation Checklist (v2 / Single Page)

Use this during implementation. The Word document is the frozen reference.

PR-0 — Preparation

- Create branch: feature/map-upgrades (or feature/)
- Verify FleetMapScreen loads and markers are visible

PR-1 — Pro Marker System (Mandatory First)

- Normalize types: latitude/longitude must be float (normalize if string), last_location_time must be ISO8601
- Heading: if null → bearing fallback; clamp result to 0–360
- Smoothing rule: apply only if delta < 30m AND timestamp delta < 1s; otherwise skip (no ghost/teleport smoothing)
- Follow mode: cancels on user drag; re-enables when user re-selects the same vehicle; keep zoom fixed (optional soft zoom)
- Selected marker stays on top (zIndex + render order)
- Performance: update only vehicles with changed updatedAtISO OR >1m movement

PR-2 — Vehicle Track (History)

- Backend: GET /api/vehicles/:id/track?from&to; (time, lat, lng, speed, heading, accuracy)
- If accuracy is null, default on client (e.g., 50m)
- Polyline + start/end markers; downsample to max ~500–800 points (time-thinning or Douglas-Peucker)
- Track playback MVP (optional)

PR-3 — Geofence Layer + Editor (MVP)

- Render geofences (Polygon/Circle) on map
- Editor validation: polygon >= 3 points; circle radius min 10m / max 5km
- Prevent/validate self-intersecting polygons (at least backend validation); show validation errors in UI

PR-4 — Geofence Violations → Alerts

- Violation timestamp must equal location_time (avoid server_time drift)
- Create alerts using correct DB column name: is_resolved
- Frontend: show violation markers + vehicle alert badge

Final Consistency Check (Do Not Skip)

- Track polyline direction is correct (start → end)
- Follow mode + geofence layer performance tested together
- Heading rotation tested on both iOS and Android
- Backend ↔ Frontend field names and DB columns match exactly (especially is_resolved)